

2 The eye has sensory receptors that convert light energy into nerve impulses. The ability of the human eye to detect separate stimuli one after another has been studied.

- The eye is able to distinguish separate stimuli from a flickering light up to a threshold frequency of about 16 flickers per second. This is referred to as a frequency of 16 hertz (Hz). 1 Hz = 1 flicker per second.
- Once the threshold frequency is reached, the eye can no longer distinguish the separate stimuli so they are detected as one continuous stimulus.
- The frequency at which this happens is called the critical flicker fusion threshold (CFFT).

To find a person's CFFT, a test is carried out.

- The person looks into a binocular eyepiece at two white lights that flicker simultaneously.
- The flickering frequency is started at 4 Hz and is gradually increased at a constant rate.
- When the person first notices that the flickering stimulus appears to have become constant they press a button, which gives their CFFT.
- This is repeated to give ten values for each person.

An investigation was carried out into the effect of alcohol on the CFFT of thirteen females aged from 18 to 35 years old. Before starting the investigation, the women were given time to become familiar with the equipment. The investigation was then carried out over two days.

Day 1 – the women were allowed 5 minutes to drink 250 cm<sup>3</sup> of pure orange juice. The CFFT was tested 40 minutes later.

Day 2 – the women were allowed 5 minutes to drink 250 cm<sup>3</sup> of orange juice containing 16 mg of alcohol. The CFFT was tested 40 minutes later.

The reason for carrying out the test 40 minutes after drinking alcohol is to allow time for the alcohol to reach its peak concentration in the blood.

Fig. 2.1 shows the results of this investigation.

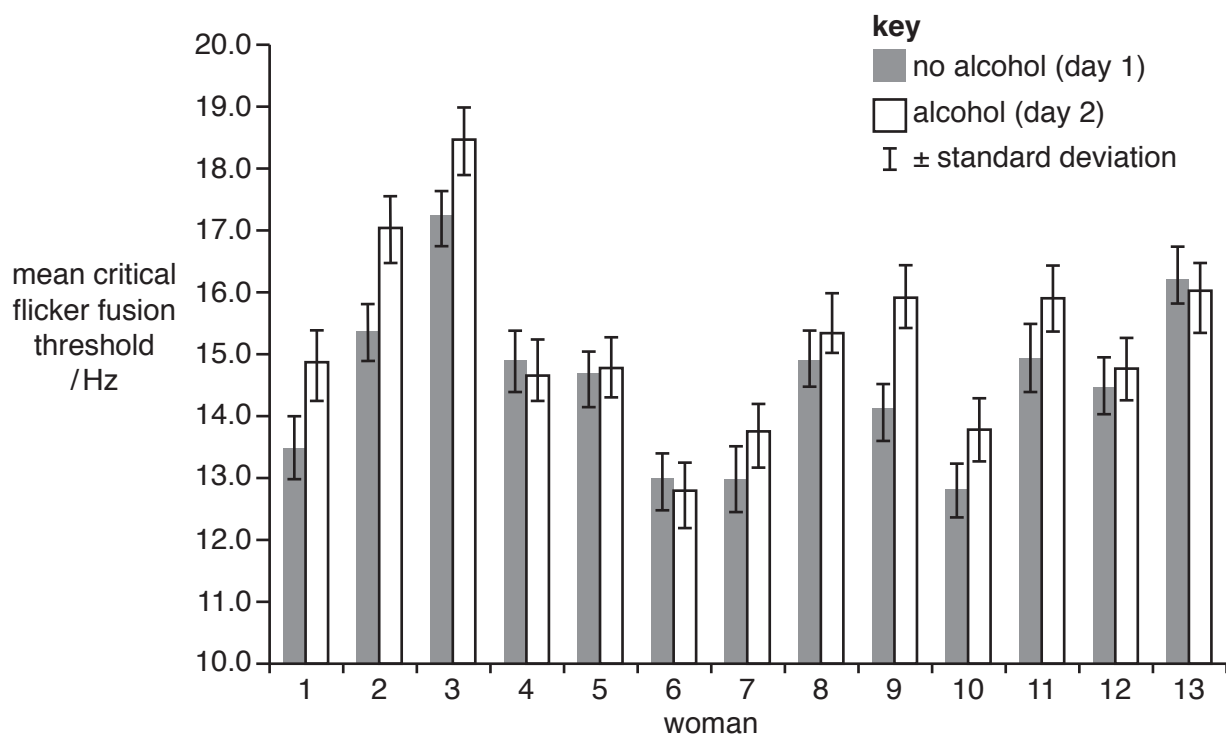


Fig. 2.1

(a) Identify **two** key variables that have been controlled in this investigation.

- 1 .....
- 2 ..... [2]

(b) Explain why the investigators calculated the standard deviation of the results for each person.

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(c) Identify **three** features about the CFFT from the data shown in Fig. 2.1.

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(d) Suggest **three** reasons why the results shown in Fig. 2.1 may **not** be representative of the whole human population.

- 1 .....
- .....
- 2 .....
- .....
- 3 .....
- .....
- ..... [3]

[Total: 10]